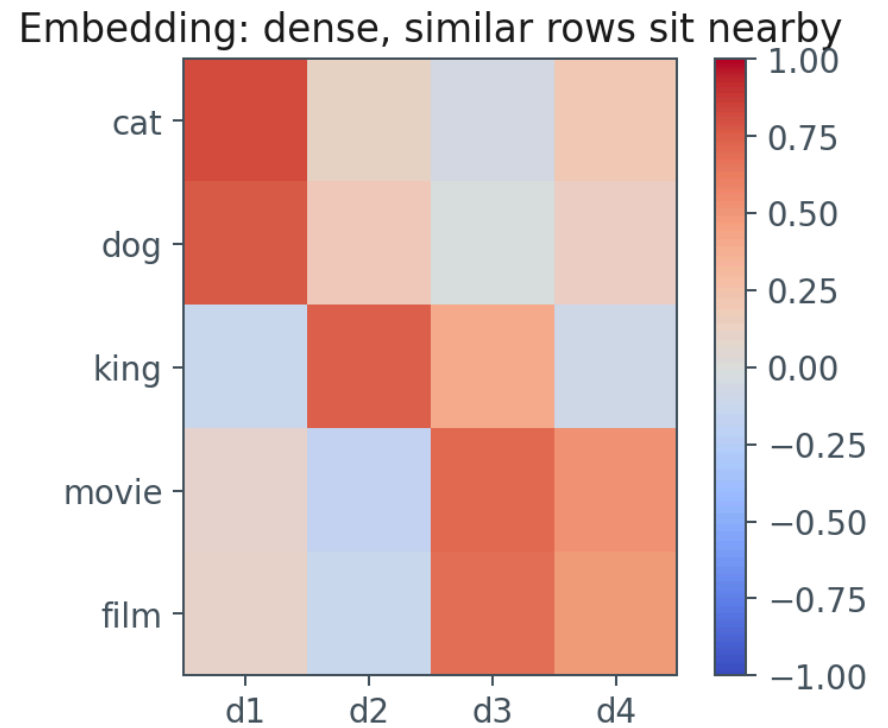
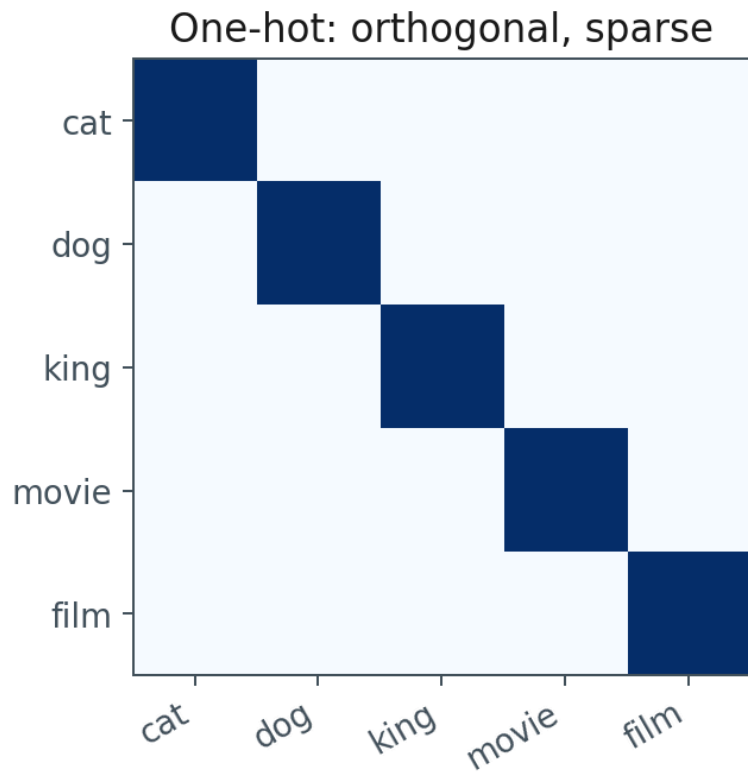


Learning goals for this block

- Separate distributional, distributed, and embedding.
- Write skip-gram $P(w_o \mid w_c)$ and a $V = 3$ softmax.
- Use cosine; run the Lab 1 signed-projection bias probe.
- Report extrinsic numbers in a project; use intrinsic to debug.

One-hot is a bad geometry

- film and movie are orthogonal. New words have no coordinate.
- Distributed = short, dense. Similar usage → similar vectors.
- An embedding is the map from a token id into that space.

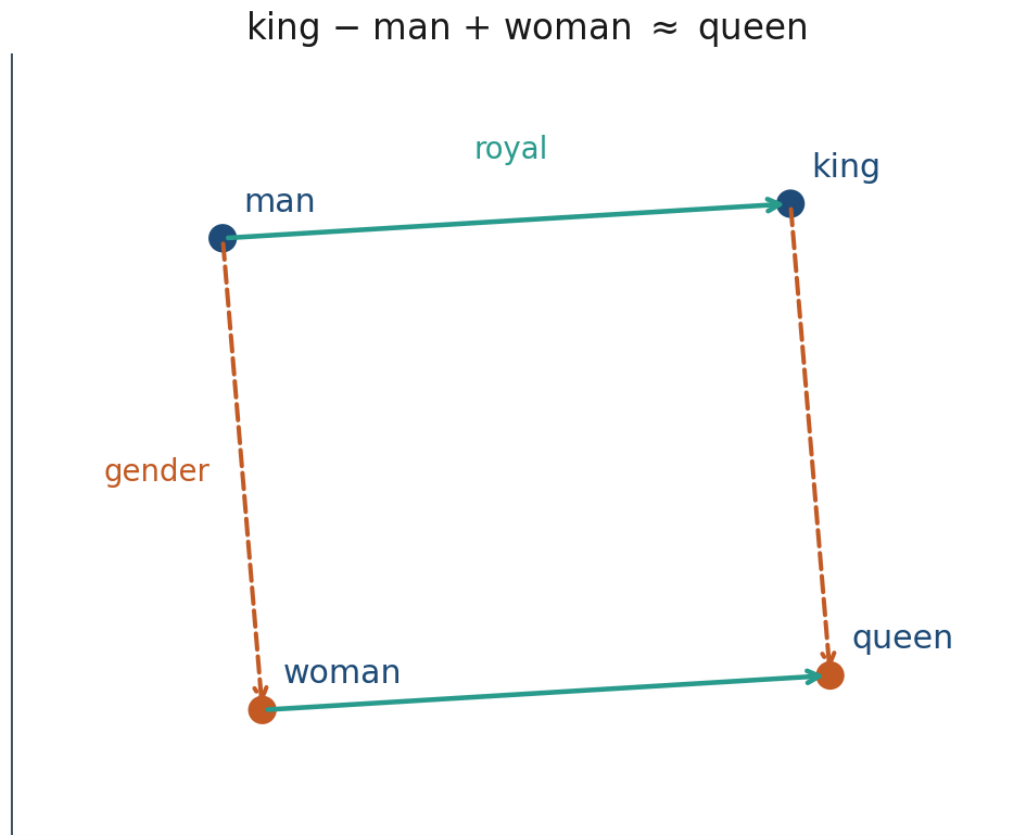


Three words that are not synonyms

- Distributional: similar contexts, similar meaning (Harris; Jurafsky Ch. 5).
- Distributed: the code is a dense vector, not a one-hot.
- Embedding: the lookup, `nn.Embedding`, row of a $V \times d$ table.
- Mixing these three is the most common exam slip this week.

Direction can mean something

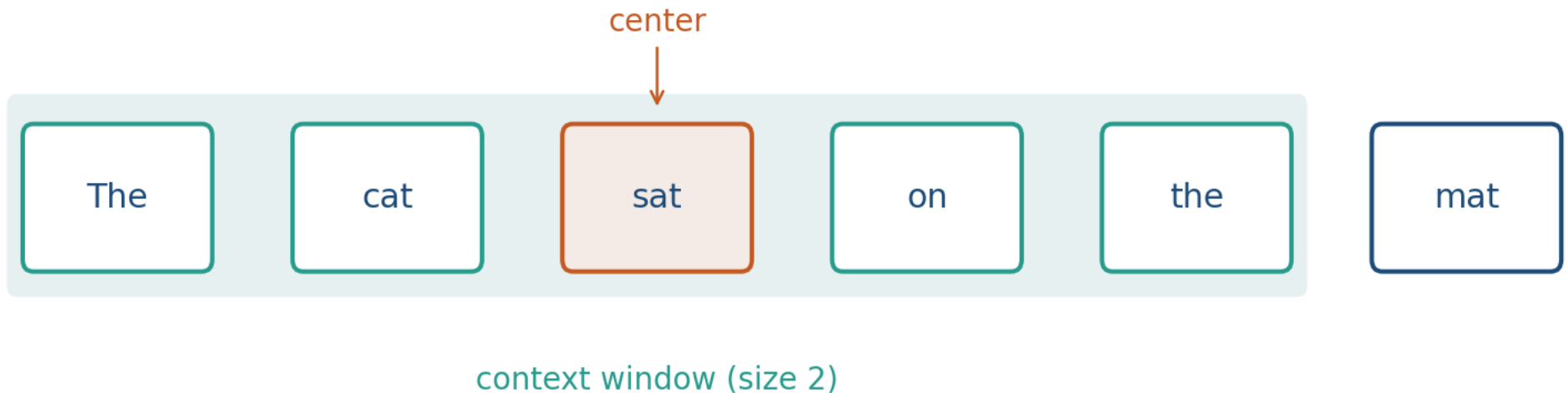
- king – man + woman $\approx$ queen.
- Cosine: $u^T v / (\|u\| \|v\|)$. Length tracks frequency.
- Nearest neighbors are a debug tool, not a project metric.



Skip-gram learns the map

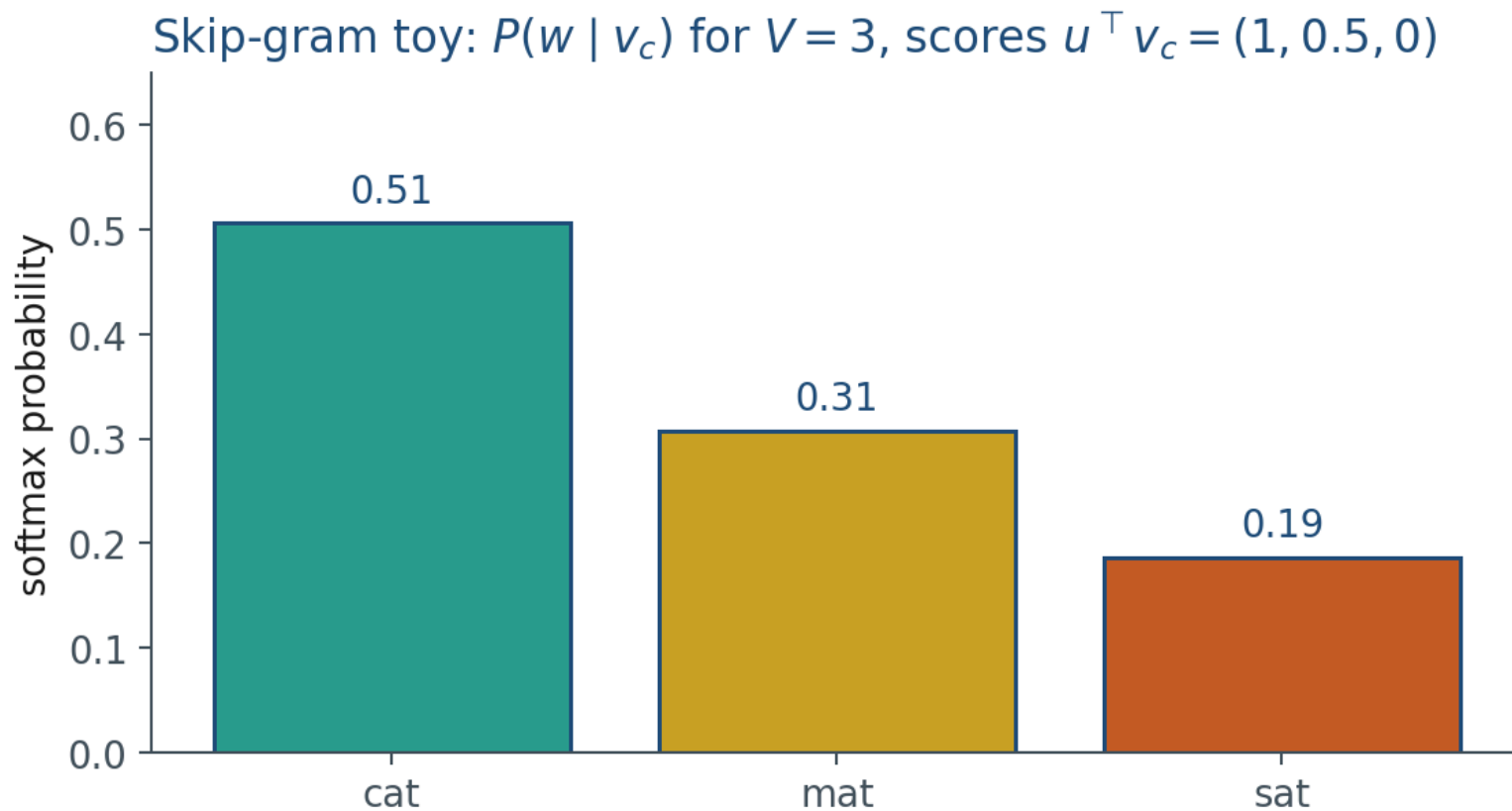
- Fake task: from the center, predict each neighbor.
- CBOW: from the neighbors, predict the center.
- Logits are $z_w = u_w^\top v_c$. Softmax over the vocabulary.

Skip-gram: predict each neighbor from the center word



Skip-gram softmax, $V = 3$

- $P(w_o | w_c) = \exp(u_o^\top v_c) / \sum_w \exp(u_w^\top v_c)$.
- Scores (1, 0.5, 0) for cat, mat, sat $\Rightarrow p \approx (0.51, 0.31, 0.19)$.
- Mikolov 2013a §§1–3. Negative sampling is cited, not derived.



Keep the lookup; throw the softmax

- `nn.Embedding(V, d)` is a table. One-hot $\times$ matrix is the same map.
- Word2Vec keeps v_w . It throws away the output softmax.
- A transformer keeps transforming those vectors instead.

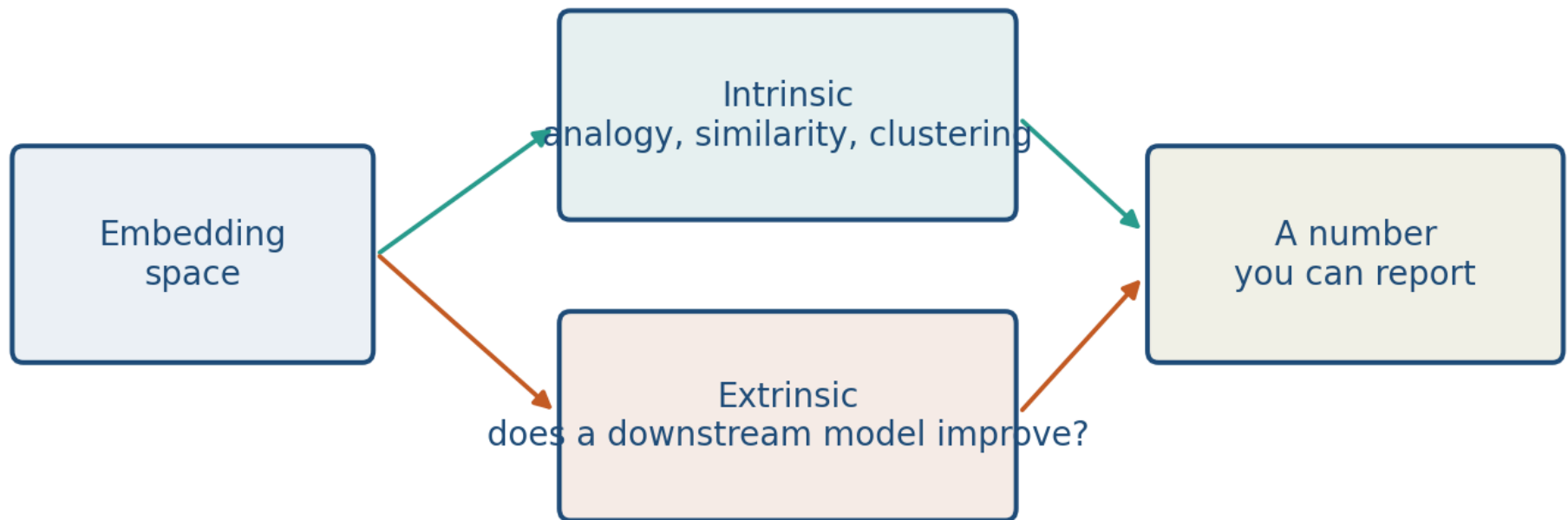
Counting vs. predicting

- Classical vector semantics can start from a co-occurrence matrix.
- Skip-gram predicts context instead of counting it.
- Both are distributional. We use the predictive story because it is the LLM loop.
- No GloVe SVD homework this week. Optional later: CS224N A1.

Intrinsic vs. extrinsic

- Intrinsic: analogies, similarity, clustering of the space.
- Extrinsic: does a downstream model improve?
- They need not agree. Projects report extrinsic.

Two ways to score embeddings. They need not agree.



Bias is also geometry

- Offset $o = \vec{\text{he}} - \vec{\text{she}}$.
- Score $v^\top o / \|o\|$. Larger sits closer to he.
- Lab 1: constructed 2-D CSV first, then the same probe on a real model later.
- Debiasing is incomplete. Report the probe when you rank people or jobs.

Geometry can encode stereotypes as well as meaning

